Problem Chosen	2024	Team Control Number
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	Summary Sheet	

Coastal Insurance with Solvency, Affordability, and Heritage Constraints

Summary

Property insurance remains sustainable only when a price can fund expected claims and catastrophic claims while enough owners can afford it. We connect these requirements to physical mitigation, development choices, and the preservation of St Mark's Basilica in Venice.

Official 2002–2019 observations generate flood-loss envelopes in each local height datum, bracketing annual-maximum damage and complete repair between tides. A demand model selects the lowest price covering expenses, capital and a common full-limit claim. Preservation keeps historic integrity separate from insurable restoration.

For the declared residential financial scenario, Charleston's 0.9 m floor is rejected even at the unshifted baseline. A 1.2 m floor admits a 0.90% annual premium, but loses viability under a 0.30 m water-level shift; raising it to 1.5 m restores viability. In Venice, a 1.7 m floor admits 2.37% at baseline, whereas a 0.15 m shift requires a higher floor or different financial terms. These are conditional single-year archetype quotes.

Single-year solvency does not guarantee renewal. A legal full-cap claim leaves the Venetian insurer only $11/180$ of V, with no feasible next-year quote in the same family. We therefore test eight three-year zero/full-cap paths and price a separately funded, loss-absorbing capital commitment. At a stated 2% commitment fee, Charleston requires 1.067%, 733 clients and $109.95V$ of sufficient outside three-year capital. Both tested Venetian cases then fail the original affordability scenario. The paths carry no assigned probabilities.

Insurance alone does not justify construction. Raising a Charleston building without its access corridor fails the service test. With the replenishment fee included, all three designs fail at a 6% annual use margin. Raising site and access reaches financial break-even at 8.45%, conditional on funded renewal throughout the 30-year appraisal; a tested three-year commitment alone does not secure it.

For St Mark's, height censoring permits 14–16 of 18 years to cross 1.40 m after a 0.30 m shift. An eight-state linear program propagates these intervals with arbitrary failure dependence. Under declared reliability scenarios, audited protection reduces the worst annual intrusion opportunity from 6.0% to 1.5%. The EUR 195,469 single-year quote fits EUR 200,000, but a 2% replenishment fee raises it to EUR 207,781 and requires EUR 1.4775 m outside capital over three years. The original annual budget therefore fails this renewal alternative. We propose an immediate EUR 250,000 audit, conditional EUR 1.2 m local works and EUR 80,000 annual operations; any expanded coverage budget and capital provider require explicit approval and verified terms.

Keywords: catastrophe insurance; common shock; demand; flood adaptation; heritage; dependence ambiguity.

Contents

1 The decision is a joint feasibility problem . . . 3

2 Evidence, common assumptions, and their uses . . . 3

2.1 Two local datasets and a historical boundary . . . 3

2.2 Why the assumptions matter . . . 4

3 From high water to a contract loss . . . 5

3.1 Flood depth and annual loss envelopes . . . 5

3.2 Binned data and unresolved observations . . . 5

4 A sustainable quote needs customers and capital . . . 6

4.1 The joint cash-flow constraints . . . 6

4.2 An exact decision within the declared policy family . . . 7

4.3 Solvency must be checked again at renewal . . . 7

5 Two continents, with different local thresholds . . . 9

5.1 Regional quote decisions . . . 9

5.2 Challenges that can reverse a decision . . . 10

6 Where, how, and whether to build . . . 11

7 Preservation without an arbitrary cultural price . . . 13

7.1 Which buildings deserve protection? . . . 13

7.2 How much preservation is justified? . . . 13

8 St Mark's Basilica: significance and protection . . . 14

8.1 Significance and existing measures . . . 14

8.2 Dependent failure can reverse the apparent best measure . . . 15

9 Insurance and a staged preservation decision . . . 16

9.1 The restoration allowance is not the monument's value . . . 16

9.2 A renewal promise requires additional funding . . . 17

9.3 Recommended extent, costs, and decision gates . . . 18

10 What is verified, and what remains to be measured . . . 19

Letter to the Venice community . . . 21

Report on Use of AI . . . 22

1 The decision is a joint feasibility problem

The insurer's problem is not simply to estimate the next disaster. A small expected loss can coexist with a large correlated claim, and an actuarially adequate price can eliminate the customer base that pays fixed costs. The community's problem is wider still: an insurable building can occupy a site with unsafe access, and an uninsurable monument can retain irreplaceable value.

We answer the original tasks in four linked decisions. First, determine whether an insurer can offer a viable flood contract and identify the changes an owner can make. Second, demonstrate the method in Charleston, South Carolina, in North America, and Venice, Italy, in Europe. Third, connect the contract to development cash flows and local access. Fourth, select the extent of preservation and apply it to St Mark's Basilica, followed by a one-page community letter.

The peril studied quantitatively is coastal high water. Hurricane winds, rainfall drainage, fire, liability, and business interruption require their own claim modules. Their omission does not imply that a multi-peril property policy is safe. Coastal flood coverage is a useful starting point because tide observations identify an actual physical hazard on both continents, while the same monetary loss model can be subjected to alternative vulnerability and contract assumptions.

The resulting decision has three possible interpretations: a feasible quote within the stated financial and physical domain; rejection within that domain; or an unresolved decision when a required input is missing. Rejection of a particular contract, insurer balance sheet, or design does not prove that every possible insurer or protective measure must reject the area.

2 Evidence, common assumptions, and their uses

2.1 Two local datasets and a historical boundary

NOAA station 8665530, Charleston, supplies hourly water levels in metres relative to mean higher high water (MHHW). We downloaded one annual original for each year 2002–2019 through the official Data API[1]. We retain all supplied values, identify local hourly maxima, record missing observations, and independently read each annual maximum from the original files. A water level at the harbour gauge is not a mapped flood depth at every property.

For Venice, ISPRA supplies annual counts above 80 cm and 110 cm at Punta della Salute, while the municipality supplies annual severity bins and a list of exceptional heights[2, 3, 4]. Heights are relative to the 1897 Zero Mareografico di Punta della Salute (ZMPS). The city archive describes a 1–2 cm possible difference between its Canal Grande instrument and the nearby ISPRA instrument[5]. We keep the original datasets and their differences; we do not assume that every listed high tide is an independent storm.

In the same records, Charleston's annual hourly-peak crossings above the study threshold 0.50 m MHHW average 4.33 in 2002–2010 and 12.22 in 2011–2019. Venice's annual tides at or above 1.10 m ZMPS average 6.67 and 8.56 in the same two periods. These are within-gauge descriptive crossing counts, not comparable storm counts, significant trend tests, or a causal estimate of climate change. They show why changing physical exposure belongs in the

underwriting decision.

We use the common 2002–2019 window because Venice's MOSE intervention changes observed lagoon extremes from 2020 onward. Post-2020 marine and lagoon observations are different physical quantities[2]. Sources were accessed on 13 October 2026. Thus the evidence is a 2026 reconstruction of the 2024 question, with a pre-intervention hazard window. Later infrastructure information is used only to avoid proposing work that already exists. It is not inserted into the historical hazard record.

2.2 Why the assumptions matter

Table 1. Common assumptions and their effects

Assumption	Reason and use	Failure or required check
Local datum is retained	Flood depth requires water and floor heights in the same vertical frame.	Survey and gauge-to-site routing must precede an actual quote.
Annual replay of 18 years	Preserves observed concentration within a year without fitting an unsupported rare tail.	This empirical mean is not a dated climate forecast or a 99% guarantee.
Repair between tide episodes	Gives an upper loss scenario; no repair gives an annual-maximum lower scenario.	Duration, salt and repeat damage require claims and repair records.
Common local weather	All policies in a homogeneous archetype can lose together.	Spatial diversification requires an independently supported joint model.
Explicit demand and cost scenario	Tests the profitability–affordability mechanism.	Customer surveys, real expenses and reinsurance quotes must replace scenario inputs.
Restoration differs from heritage	Money can repair damage without recreating authenticity or identity.	Community priorities and conservation approval are required.

The first-floor heights, vulnerability slope, capital, expenses, customer budgets, failure rates, and proposed project budgets below are declared scenarios. None is represented as a measured local effect. The assumptions have different roles: datum retention defines a valid subtraction; repair timing defines a loss envelope; common weather protects the capital calculation; financial scenarios define a conditional feasible set. Making the list longer would not make those assumptions true.

A globally consistent depth–damage database developed by Huizinga, de Moel and Szewczyk relates fractional damage to depth and discusses adjustments for local circumstances[6]. It supports the modelling architecture, but does not calibrate our coastal

archetypes or an ancient basilica. We therefore use an explicit linear vulnerability scenario and attack its slope, rather than claim that a regional curve validates a particular building.

3 From high water to a contract loss

3.1 Flood depth and annual loss envelopes

Let H_{ye} denote the height of exposure episode e in year y, z the vulnerable floor, Δ a water-level shift, and V an insured reconstruction-and-contents allowance. All heights at a site use its local datum. The declared vulnerability is

$$d_{ye} = \max(0, H_{ye} + \Delta - z), \qquad D(d) = \min(1, \beta d), \quad \beta = 0.5\ \mathrm{m}^{-1}. \tag{1}$$

The value V excludes land and is not a property-sale price. We vary β between 0.25 and 1.0 per metre. This is a structural sensitivity range, not a confidence interval fitted to claims.

An event deductible aV with $a = 0.02$ and annual aggregate limit cV with $c = 0.5$ define the contract. In units of V, the annual single-maximum and complete-repair claim amounts are

$$\ell_y^- = \min\left(c, \max_e [D(d_{ye}) - a]_+\right), \qquad \ell_y^+ = \min\left(c, \sum_e [D(d_{ye}) - a]_+\right). \tag{2}$$

Here a and c are dimensionless fractions in the equation. In a year with only one loss episode the endpoints coincide. In a year with repeated losses, the upper endpoint permits a fresh deductible and a full repair before the next tide. That reset is deliberately adverse; two tidal peaks within one storm are not claimed to be two observed insurance claims. The cap prevents repeated events from silently creating an unlimited annual promise. We compute the corresponding gross physical repair scenario $g_y^+ = \min(1, \sum_e D(d_{ye}))$ and keep it distinct from insurance indemnity.

Equation(2) brackets the stated reset/no-reset models. It does not certify all salt-deterioration mechanisms or unobserved peaks. Charleston's hourly peaks may be lower than between-hour maxima. A further 50 mm height challenge checks the decisions most exposed to sampling resolution; it is not a certified interpolation error bound.

3.2 Binned data and unresolved observations

For Venice's lower envelope we use an exact documented annual maximum when available, otherwise the highest occupied bin's lower endpoint. This retains water-height ambiguity in addition to repair timing. For its upper envelope, a bin $[h_j^-, h_j^+)$ with count n_{yj} contributes at most $n_{yj} D(h_j^+ + \Delta - z)$ to the complete-repair calculation because D is nondecreasing. The open upper endpoint gives a conservative supremum. The uppermost historical bin is assigned 1.94 m, the city's recorded historical extreme; this is a specified challenge endpoint, not a fitted return level. We have exact annual maxima whenever they reached 1.40 m, including 1.87 m in 2019[4].

The count files disagree in a few years. In 2019, the municipality's six-bin total is 26 while the ISPRA count at or above 110 cm is 28. In 2012 the six-bin total is 8 while the ISPRA maximum count is 7; the municipality explicitly includes certain high minima[3, 2]. We do

not infer an undocumented explanation for every discrepancy. The upper scenario retains the municipality's episodes and adds any positive difference as unresolved episodes at 1.40 m. This is an adverse ambiguity envelope over the low-class discrepancy, not an exact reconstructed event chronology. The 2019 official peak list is separately retained as a source check.

Omitted observations below 110 cm cannot damage the residential archetypes in our table because $z - \Delta \geq 1.10$m. That condition is essential. The residential calculation is not reused for the Basilica's low interior. Its protection model uses annual intrusion states instead.

The scenario shifts $\Delta \in \{0, 0.15, 0.30\}$m increase the frequency of crossings even when the number of weather systems is unchanged. They are stress increments, not an asserted forecast for a particular year. Future weather can exceed this 18-year record; the solvency check therefore uses the full contractual loss cap instead of the largest historical fitted quantile.

4 A sustainable quote needs customers and capital

4.1 The joint cash-flow constraints

Let n identical policies be enrolled in one local archetype and let pV be the annual premium per policy. The local year loss is $nV\ell_y$, not the sum of independent draws. Let r be the share retained by the insurer. We assume a fully collectable quota-share reinsurance contract whose quoted cost per policy, in units of V, is

$$R_r = (1 - r)(1.25\mu + 0.001), \qquad \mu = 18^{-1} \sum_y \ell_y^+. \tag{3}$$

The 25% load and standby charge are financial scenarios; actual reinsurance bids are required. Changing the retained share does not remove the total owner's loss. Counterparty default or simultaneous regional reinsurance failure would require another capital stress.

The demonstration balance sheet has $N = 1000$ potential clients and available capital K. Its dimensionless capital and fixed overhead are $k = K/V = 50$ and $f = F/V = 0.5$. Per-policy administration is bV, with $b = 0.0005$; capital opportunity cost is $\rho = 0.08$, and premium expense share is $e = 0.20$. We include ρK in annual costs rather than treat capital as free. Expected operating surplus and one-year stress capital are

$$\Pi/V = n\{(1 - e)p - r\mu - R_r - b\} - f - \rho k, \tag{4}$$

$$W/V = k + n\{(1 - e)p - rc - R_r - b\} - f - \rho k. \tag{5}$$

On the right, k, f, b, p, μ and R_r are dimensionless fractions or multiples of V. A quote requires $\Pi \geq 0$ and $W \geq 0$ simultaneously. The stress is all n policies at their annual limit; it is a contract-based bound conditional on the reinsurance promise. Neither a historical loss average nor a large number of local policies justifies dropping this stress.

The demand schedule consists of N owners with willingness to pay $w_j/V = A(1 - j/N)$, $j = 1, \ldots, N$, where the dimensionless annual budget ceiling is $A = 0.04$. It is a transparent affordability scenario, not an income estimate for either city. A firm may cap enrollment at n and select the n eligible applicants. Enrollment is possible at price p only if

$$p \leq A(1 - n/N). \tag{6}$$

All capital K in this single-portfolio scenario is assigned to the portfolio and bears the stated annual opportunity cost. Another insurer can choose a different dedicated allocation; total available corporate funds alone do not force that allocation.

This mechanism captures the original dilemma: fewer clients reduce claims but raise fixed cost per client. All clients pay the same archetype price; household assistance is considered separately from risk pricing.

4.2 An exact decision within the declared policy family

For each integer $n \in \{1, \dots, N\}$ and retained share $r \in \{0.10, 0.25, 0.50, 1\}$, define

$$p_{nr} = \max\left\{0, \frac{r\mu + R_r + b + (f + \rho k)/n}{1 - e}, \frac{rc + R_r + b + (f + \rho k - k)/n}{1 - e}\right\}. \tag{7}$$

The firm should underwrite if at least one pair satisfies $p_{nr} \le A(1-n/N)$. Among feasible pairs, we choose the smallest premium and then the largest enrollment at a tie. This is a minimum sustainable quote, not the firm's profit-maximising monopoly price.

Why the test is sufficient and necessary in this family. For fixed n, r, both cash-flow constraints are increasing affine functions of p. Their smallest common feasible price is exactly(7). If this price exceeds the largest price that admits n applicants, no price works for that pair. If it does not, the pair is a constructive feasible witness. Enumerating all 4000 pairs proves the result over the declared enrollment and retention choices. It does not exclude other insurers, demand curves, deductibles, subsidies, or reinsurance structures.

A feasible policy is never confused with a generally necessary resource vector. The reported capital and reinsurance mix is one explicit balance sheet. Another mix may work. Nor does nonnegative expected profit prove solvency: for example, with $n = 1000$, $K = 50V$, $r = 0.25$, and a 0.5 V annual claim, the retained catastrophe alone is 125 V. Pricing an almost loss-free history at approximately its expenses can leave a 75 V capital shortfall before overhead. This analytic counterexample is why the second term in(7) is necessary.

The same result explains the owner's influence. Raising the vulnerable floor lowers d and μ; documented water-resistant restoration can lower β; verified district protection can change the height distribution. A higher deductible may reduce claim expense but also shifts loss to an owner who must still afford repairs. Mitigation credit should be conditional on survey and performance evidence, with inspections at renewal. An undocumented promise to close a barrier receives no quantified credit.

4.3 Solvency must be checked again at renewal

Equations(7) and(6) certify a single annual contract. For annual realized claim fraction $\ell_t \in [0, c]$, let i_t be an outside capital injection and d_t a positive surplus transfer. The normalized capital recursion is

$$\widetilde{k}_{t+1} = k_t + n_t\{(1 - e)p_t - r_t\ell_t - R_t - b\} - f - \rho k_t - \gamma h_t, \quad k_{t+1} = \widetilde{k}_{t+1} + i_t - d_t. \tag{8}$$

Here h_t is the outside commitment in units of V, and γh_t is its annual fee. In the no-facility case, $\gamma = i_t = d_t = 0$, and the insurer re-solves the annual quote at its actual next-year capital. It cannot assume that capital returns to $50V$.

A continuation counterexample. Venice's 1.7 m baseline quote has $n = 404$, $r = 0.25$, and $\mu = 1/180$. A full annual $0.5V$ claim leaves

$$k_2 = 50 - 404(0.25)(0.5 - 1/180) = 11/180. \tag{9}$$

It is solvent in that year, but independent enumeration finds no affordable second-year quote in the declared family. Charleston's zero-indemnity archetype similarly falls from 50 to $11.3V$ after a full-cap year. Its next quote admits only 226 clients; a second consecutive full-cap year leaves zero capital and no third-year quote. Historical zero indemnity therefore does not secure a long insurance promise. These examples test the minimum-price adaptive policy, not every possible dynamic pricing or financing strategy.

A priced, finite commitment. As an explicit alternative, a separate provider prefunds loss-absorbing capital for three years. It promises non-repayable capital transfers; there is no loan balance or additional dividend obligation during this window. The provider receives the stated fee and positive surplus, bears called capital losses, and recovers uncalled funds at expiry. This is a conditional grant/equity contract, not a demonstrated market offer. A loan would require a debt and repayment state and cannot replace it silently.

For a fixed portfolio at target capital $k = 50$, choose a sufficient annual commitment

$$h = nr(c - \mu), \quad i_t = [k - \widetilde{k}_{t+1}]_+, \quad d_t = [\widetilde{k}_{t+1} - k]_+. \tag{10}$$

The provider prefunds $3hV$, separately from the insurer's initial $50V$. We set $\gamma = 0.02$ as an unverified annual fee scenario. Add $\gamma r(c - \mu)$ to the per-policy costs in both floors of(7), then enumerate enrollment and retention again. The insurer must pay every claim before an injection arrives: the modified full-cap solvency floor is retained.

Why this commitment suffices. The expected surplus after its fee is nonnegative. Writing Π_γ for expected surplus after the commitment fee, pre-refill capital at full loss is $k + \Pi_\gamma/V - h$, so the required refill is at most h. Smaller claims require no larger refill; surplus transfers reset capital to k. Induction supports all three years with at most $3hV$ calls. This is a sufficient commitment for this policy, not a generally necessary capital vector or an infinite-horizon guarantee.

Table 2. Three-year renewal and capital commitments

Case	Fee-priced quote	Clients	3-year fund	Covered paths
Charleston 1.2 m, baseline	1.067%	733	$109.95V$	8/8
Charleston 1.5 m, 0.30 m shift	1.067%	733	$109.95V$	8/8
Venice 1.7 m, baseline	Reject	–	–	0/8
Venice 2.0 m, 0.30 m shift	Reject	–	–	0/8
Basilica, expanded budget	EUR 207,781	1	EUR 1,477,500	8/8

Each path has three zero/full-contract-limit states. Counts are exhaustive stress cases, not probabilities. Basilica uses an explicit EUR 208,000 budget alternative; its original EUR 200,000 budget fails.

The independent validator checks all 40 three-year paths. Without the facility, only six of eight Charleston paths and two of eight Venetian or Basilica paths retain some quote in each

year; customer enrollment can shrink. The funded Charleston policy has $h = 36.65$, annual fee $0.733V$, and pre-refill capital $13.35V$ even on three consecutive full-cap years. The provider then pays $109.95V$ and receives $2.199V$ fees, a $107.751V$ net cash loss. Capital arrives from a named external funding role, not from the model creating free reserves. No evidence shows a provider accepts these terms. A higher fee, failure to prefund, or counterparty default can defeat the alternative. Venice's two annual-feasible cases fail even with this 2% fee in the original demand scenario; other budgets, retention domains, subsidies or insurers remain possible.

5 Two continents, with different local thresholds

5.1 Regional quote decisions

Table 3. Charleston annual residential quote scenarios

Floor (m)	Shift (m)	Loss interval (‰)	Premium (‰)	Clients	Retained
0.9	0.00	1.06–1.06	Reject	–	–
1.2	0.00	0.00–0.00	0.90	774	10%
1.5	0.00	0.00–0.00	0.90	774	10%
0.9	0.15	2.28–2.28	Reject	–	–
1.2	0.15	0.34–0.34	1.66	585	10%
1.5	0.15	0.00–0.00	0.90	774	10%
0.9	0.30	4.38–5.97	Reject	–	–
1.2	0.30	1.06–1.06	Reject	–	–
1.5	0.30	0.00–0.00	0.90	774	10%

Heights are metres above MHHW. Loss and premium are annual fractions of V.

Table 4. Venice annual residential quote scenarios

Floor (m)	Shift (m)	Loss interval (‰)	Premium (‰)	Clients	Retained
1.4	0.00	2.11–4.83	Reject	–	–
1.7	0.00	0.36–0.56	2.37	404	25%
2.0	0.00	0.00–0.00	0.90	774	10%
1.4	0.15	5.47–12.72	Reject	–	–
1.7	0.15	0.78–1.08	Reject	–	–
2.0	0.15	0.00–0.14	1.19	703	10%
1.4	0.30	10.31–32.06	Reject	–	–
1.7	0.30	2.11–4.83	Reject	–	–
2.0	0.30	0.36–0.56	2.37	404	25%

Heights are metres above ZMPS 1897; pre-MOSE replay. Loss and premium are annual fractions of V.

The tables apply identical financial terms, not identical absolute tide elevations. At Charleston's 1.2 m baseline floor, the record generates no indemnity above the deductible, yet

the minimum premium remains 0.90% because operating costs, capital and reinsurance are not free. Raising Charleston's 0.9 m floor by 0.3 m changes baseline rejection to a viable quote. Under the 0.30 m stress shift, another 0.3 m of floor height is needed within the tested choices. A policy owner can therefore influence underwriting through verified physical adaptation, rather than merely negotiate away the flood risk.

Venice's 1.4 m floor has a 4.83% upper annual claim fraction even before the shift and fails the market–capital test. The 1.7 m floor is feasible at baseline, with 404 policies, 25% retention and a 2.37% premium. With a 0.15 m shift it is rejected. A 2.0 m floor restores feasibility at 1.19%. Under the 0.30 m shift, the 2.0 m floor takes the earlier 1.7 m-floor position: 2.37%, 404 policies. This translation follows directly from $H + \Delta - z$ and provides an internal dimensional check.

These results identify conditional thresholds for a flood contract. They do not recommend elevating every real Venetian residence, nor classify every property in Charleston identically. A historic structure, a building with a basement, and a dwelling served by an unsafe road need different inputs. Real implementation creates finer risk classes, changes the demand schedule, and aggregates their loss on the same annual weather state.

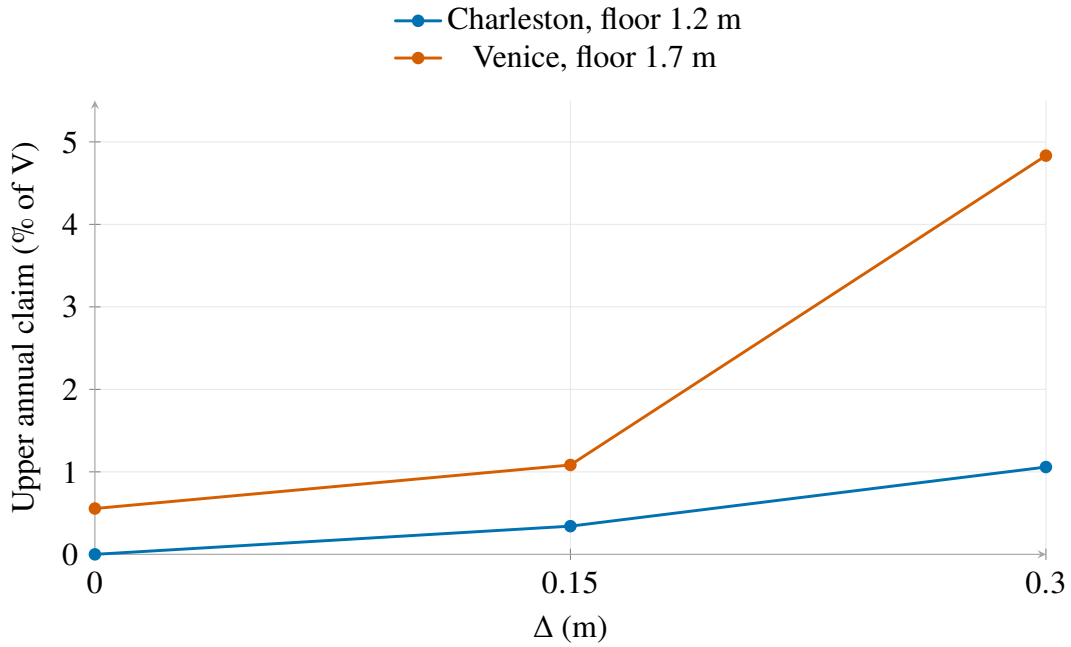


Figure 1. Water-level shifts and claim exposure. Points are declared stress scenarios on the same historical record, connected only to guide the eye. Local floor datums differ; monetary fractions are comparable.

5.2 Challenges that can reverse a decision

At a 0.30 m shift, changing Charleston's 1.2 m-floor vulnerability slope from 0.5 to 0.25 per metre lowers the upper claim fraction from 1.06% to 0.42% and changes rejection to a feasible quote. The larger slope 1.0 remains rejected. The decision therefore needs claims-based damage calibration; the datum check alone cannot validate vulnerability. Venice's 1.7 m floor remains rejected at all three slopes under the same shift, strengthening that conditional conclusion.

Adding 50 mm to Charleston's 0.30 m shift keeps the 0.9 m and 1.2 m floors rejected. The 1.5 m floor remains feasible but develops a 0.064% upper annual claim fraction. Hence its zero-claim baseline should not be advertised as flood-proof. This is an explicit sampling-resolution challenge, not an empirical upper error guarantee.

The maximum/sum envelope is also a genuine alternative, rather than a second name for the same model. For Charleston's 0.9 m floor at a 0.30 m shift, its endpoints are 4.38% and 5.97%. For Venice's 1.4 m floor they are 10.31% and 32.06%. Repeated tide exposure, repairs and contract aggregation can materially change claims. We use the adverse endpoint for quotes while preserving the lower endpoint for inspection and refinement.

Finally, no false geographical diversification is granted. A portfolio concentrated in one archetype has a linear catastrophe loss $nV\ell$, not a standard deviation shrinking as $\sqrt{n}$. A genuinely diversified cross-continent portfolio might require less capital, but independence between the two regions, especially through a shared reinsurer, has not been established. We retain a conservative capital bound until a joint model and counterparty terms justify the credit.

The feasibility region also depends on the joint capital–market choice. For a fixed dedicated capital allocation k, the smallest budget ceiling admitting any tested policy is

$$A_{\min}(k, \mu) = \min_{1 \le n < N,\ r} \frac{p_{nr}(k, \mu)}{1 - n/N}. \tag{11}$$

With Charleston's 1.2 m floor, a 15 V dedicated allocation admits the 4% ceiling while the 50

Table 5. Joint market and dedicated-capital thresholds

Archetype	$k = 15$	$k = 25$	$k = 50$	$k = 100$
Charleston, 1.2 m	3.53	4.04	5.12	7.11
Venice, 1.7 m	10.30	10.97	12.15	14.01
Venice, 2.0 m	2.48	2.94	3.97	5.93

Entries are $A_{\min}$ as annual percent of V, at the 0.30 m shift. Capital is a dedicated allocation, not total corporate funds.

V allocation does not. The smaller allocation serves fewer clients and bears less opportunity cost. At the other extreme, a 100 V fixed allocation adds enough cost to require a 7.11% ceiling. Choosing among allocations is therefore meaningful; more available funds need not be assigned to this portfolio. Venice's 1.7 m floor fails a 4% ceiling in all four tested allocations, whereas its 2.0 m floor is narrowly feasible at 50 V. The result turns rejection into a precise joint financial boundary rather than a general claim that an entire city is uninsurable.

6 Where, how, and whether to build

An insurer asks whether it can finance a defined promise. A developer asks whether creating the asset is socially and financially worthwhile. Insurance can transfer loss while its expense load and the community's service costs still make construction unattractive.

For site s and construction design j, let C_{sj} include the building, land, elevation, access and mitigation works. Let u_{sj} be annual use value, v_{sj} annual service cost, p_{sj} the sustainable

quote, and $g_{sj} - \mu_{sj}$ the gross repair allowance outside indemnity. The development test is

$$\mathrm{NPV}_{sj}/V = -(C_{sj}/V) + \sum_{t=1}^{T} \frac{u_{sj} - v_{sj} - p_{sj} - (g_{sj} - \mu_{sj})}{(1+i)^t}. \tag{12}$$

The price and uninsured loss are both annual fractions of V. Subtracting the gross loss and the entire premium would double count the expected indemnity; equation(12) explicitly avoids that error. An annual limit or deductible still leaves an owner with residual damage.

Construction requires a viable contract, nonnegative NPV relative to the no-build alternative, and local service feasibility. The service check is not total regional capacity. Each site needs a reachable evacuation route, emergency access, water and electricity connections, and capacity allocation in its own catchment. A site-access pair is inadmissible when its necessary corridor floods under the design stress, even if a hospital elsewhere has spare beds.

Our Charleston illustration uses $T = 30$, real discount $i = 3\%$, use margin 0.06 V/year and service cost 0.005 V/year. The low site has a 0.9 m vulnerable floor and 1.0 m access height. The raised-building option has a 1.5 m floor but retains the same access; the raised-site option has a 2.0 m floor and 2.5 m access. Upfront increments beyond a 1 V building are 0.10 V, 0.25 V and 0.35 V respectively. These are design archetypes with stated budget assumptions, not mapped parcels or bids.

The largest supplied Charleston hourly level is 1.213 m MHHW. A 0.30 m shift plus a 0.50 m routing/extreme challenge gives a 2.013 m access stress. This chosen margin is not a return-period elevation. It is a test of whether the proposed access is consistent with the physical scenario.

Table 6. Development and access decisions

Design	NPV/V at 6%	Access	NPV/V at 9%
Low site	No viable quote	Fails	No viable quote
Raised building	−0.388	Fails	0.200
Raised site and access	−0.481	Passes	0.107

The premium here is the 1.067% replenishment-priced quote, including the 2% fee on outside capital commitment, rather than the 0.902% single-year quote. Each positive NPV assumes equivalent funding and reinsurance terms are committed for all 30 years. The exhaustive capital check covers three years only; absent longer financing, the 30-year result is a conditional appraisal and cannot authorise a 30-year promise. An annual renewal option alone has no demonstrated post-catastrophe availability and requires a revised contingent cash-flow appraisal before investment.

At the stated 6% use margin we recommend no new construction among these three options. That conclusion is not a prohibition on all development. The raised site and access reaches its break-even margin at 8.45% of V annually, and a 9% margin gives $0.107V$ NPV. The raised building reaches financial break-even at 7.98%, but its access remains disqualifying. The fee reduces each previously priced development NPV by $0.03247V$. Financial viability, funded renewal and appropriate services must therefore be evaluated together.

For an actual city, replace each archetype with a georeferenced parcel, connect road and utility edges that remain usable under the same flood state, and solve a capacity-constrained allocation or network-flow problem. Surveyed floor and corridor elevations, flood routing, population demand, and agency responsibilities are required inputs. A straight-line distance to a service, or the sum of all capacities, cannot supply those edges. The land-use authority can then compare avoiding the low site, adapting it with local access improvements, or directing growth to a serviced higher site.

7 Preservation without an arbitrary cultural price

7.1 Which buildings deserve protection?

The community first constructs an inventory with four evidence fields: uniqueness and historic fabric; present community functions and substitute facilities; local economic dependence; and exposure together with practicable interventions. Statutory designation is recorded as evidence of significance, not as a numerical multiplier. A hospital, a place of worship, an archival collection, and a historic market can be significant for different reasons.

We apply three decisions in sequence. First, exclude actions that violate safety or an agreed conservation floor. Second, remove dominated options that cost more and preserve no more of the required functions or fabric. Third, compare the remaining cost–risk–integrity frontier with the community's priorities and budget. High market value does not automatically outrank low-income community use, and a popular building is not automatically replaceable.

For an ordinary building with interchangeable services, adaptation or relocation can be compared with replacement. For an irreplaceable monument, documentation preserves information but does not replace authentic materials. The preservation objective must therefore keep an integrity requirement alongside monetary cost. This is a deliberate community constraint, not a theorem proving that all landmarks can or should be saved at any cost.

7.2 How much preservation is justified?

For action a, let C_a be initial preservation cost, O_a annual operations, π_a annual probability of a damaging intrusion year, and W the community's total loss allowance per such year. W can include restoration, interruption and the community's willingness to prevent irreversible loss; it is a preference input, not a claim that heritage has a market price. Over a horizon T,

$$J_a = C_a + \sum_{t=1}^{T} \frac{O_a + \pi_a W}{(1+i)^t}, \quad \text{subject to } I_a \geq I_{\min},\ \pi_a \leq \pi_{\max}. \tag{13}$$

The integrity condition $I_a \geq I_{\min}$ is a documented pass/fail conservation review, not a fabricated precision score. A decision may use the risk target as a hard constraint even when the community cannot monetise W.

For two otherwise acceptable actions, a more protective action b is financially preferable

to baseline$a$ when

$$W \geq \frac{(C_b - C_a) + A_T(O_b - O_a)}{A_T(\pi_a - \pi_b)}, \qquad A_T = \sum_{t=1}^{T}(1+i)^{-t}, \quad \pi_b < \pi_a. \tag{14}$$

This threshold makes the preference transparent. It avoids assigning unexplained weights to cultural, historical, economic and community scores, then multiplying a ranking by replacement value. Where no action meets safety and integrity jointly, leaders should state that conflict, protect movable collections and services, and reassess the larger action set. A failure of our finite candidate list does not exclude every possible engineering action.

8 St Mark's Basilica: significance and protection

8.1 Significance and existing measures

St Mark's Basilica lies within Venice and its Lagoon, a UNESCO World Heritage property. UNESCO identifies Venice's architectural ensembles, authenticity and relationship with the lagoon, while high water is a recognised conservation pressure[7]. The Basilica is therefore an appropriate example of irreplaceable historic fabric that should not be ranked only by a property-sale value. Its worship and visitor functions also connect it to the surrounding community.

Glass and steel protection was installed in 2022; the manufacturer describes watertight access openings and the complementary role of MOSE for higher tides[8]. A 2025 municipal agreement addresses management of the glass barriers[9]. These sources establish that existing protection must be audited and maintained, not proposed as though nothing had been built. They do not certify every rear path, foundation, drainage mechanism, effective threshold or annual failure probability.

We model a vulnerable interior below 0.80 m ZMPS and a success state for MOSE that limits lagoon water to 1.00 m. The illustrative baseline local envelope is 1.10 m; proposed qualification thresholds are 1.40 m and 1.80 m. These are scenario effective thresholds across all water paths, not verified as-built wall dimensions. Every year in the historical window has an observed tide at or above 0.80 m, so failure of the local envelope presents an intrusion opportunity in every replayed year. Routine moisture deterioration remains part of the operations programme.

The documented maximum heights above 1.40 m occur in eight of the 18 years. Under the 0.30 m shift, the source intervals allow 14–16 years to exceed 1.40 m, while exactly three exceed 1.80 m. The maximum height in 2006 and 2011 lies in the 110–120 cm bin; either could equal 110 cm, so strictly crossing 140 cm after the shift is uncertain. We retain all 18 annual lower/upper height endpoints and whether an upper endpoint is open. Counts are local height crossings, not observed insurance claim counts. A 1.80 m enclosure also risks unacceptable visual or physical intervention; it is included as a comparison, not presumed conservation-approved.

8.2 Dependent failure can reverse the apparent best measure

Let L indicate an annual local failure relevant to an interior-threatening exposure, M an annual ineffective MOSE response, and X a year whose maximum height exceeds the local envelope. We evaluate an adverse annual replay that allows each failure to coincide with an available hazardous exposure. Its intrusion-opportunity event is

$$E = L \cup (M \cap X). \tag{15}$$

Let their marginal probabilities be f_L, f_M, q_X. These are annual exposure-relevant probabilities, not per-hour unavailability or per-gate failure rates. Dry-period outages do not by themselves constitute L. The replay allows the opportunity to be realised, so its upper probability bounds restoration funding within the declared scenario; its lower bound is not a lower bound on observed claims. Because an outage may be triggered by the same weather that produces the largest tide, assuming independent failures and severity is optimistic. With arbitrary dependence,

$$\max\{f_L, f_M + q_X - 1\} \le \Pr(E) \le \min\{1, f_L + \min(f_M, q_X)\}. \tag{16}$$

Proof. The intersection $M \cap X$ has probability at least $[f_M + q_X - 1]_+$ and at most $\min(f_M, q_X)$. The union must be at least each constituent's probability and at most their sum. The extremes are attainable by putting the minimum possible intersection inside L as far as its probability permits for the lower bound, or its maximum possible intersection outside L until probability one is filled for the upper bound. An independent eight-state linear program, with nonnegative probabilities and the three marginal equalities, verifies both extrema for every tabulated scenario. When height censoring gives $q_X \in [q_X^-, q_X^+]$, the lower envelope uses q_X^- and the upper uses q_X^+; monotonicity makes these endpoints sufficient. The independence value is likewise an interval; a retained point value is explicitly its upper-height-endpoint scenario.

Under full independence, the point estimate would instead be

$$\pi_{\text{ind}} = f_L + (1 - f_L) f_M q_X. \tag{17}$$

It is displayed as a competing structural assumption, not as measured reliability. ISPRA reports two lagoon exceedances of 110 cm in 2020 and one in 2024, occurring in two of the five years 2020–2024[2]. Deployment rules and operating regimes differ from our qualified success state, so these observations cannot directly estimate its annual failure marginal. They nevertheless provide no support for treating 1% as an achieved operational rate. The near 1% case is a performance condition for the illustrative private quote. The immediate recommendation is the audit and operations phase; construction and coverage commitments remain conditional.

The annual failure rates are scenario marginals. They include failure to deploy, power, leakage and loss of effective capacity. There are no operational data here proving that a real system achieves a selected rate.

For a qualified 1.40 m local envelope at the 0.30 m shift, set $f_L = 0.005$. If $f_M = 0.01$, the independence interval is 1.274–1.384%, while the sharp adverse value remains 1.50%. At $f_M = 0.05$, independence gives 4.369–4.922% and the adverse value rises to 5.50%, failing the 2% target. Increasing the envelope to 1.80 m then looks safe under independence (approximately 1.33%), yet the adverse value remains 5.50%. The 1.80 m envelope has three exact shifted

crossings, so its nominal line is unaffected by censoring. All 27 worst-dependence values remain unchanged, for two distinct reasons. In each height-censored group, the crossing-fraction lower bound is at least 10/18, above every tested regional failure marginal (at most 0.10); its upper risk bound therefore remains $f_L + f_M$. The other groups have exact, unchanged crossing fractions. In particular, the unshifted 1.80 m envelope has $q_X = 1/18 < 0.10$, so that case is unchanged through exact height information, not failure-marginal saturation. A stronger wall is insufficient when regional failures can be concentrated in the few years that overtop it. This is a concrete failure of the independent model's recommendation.

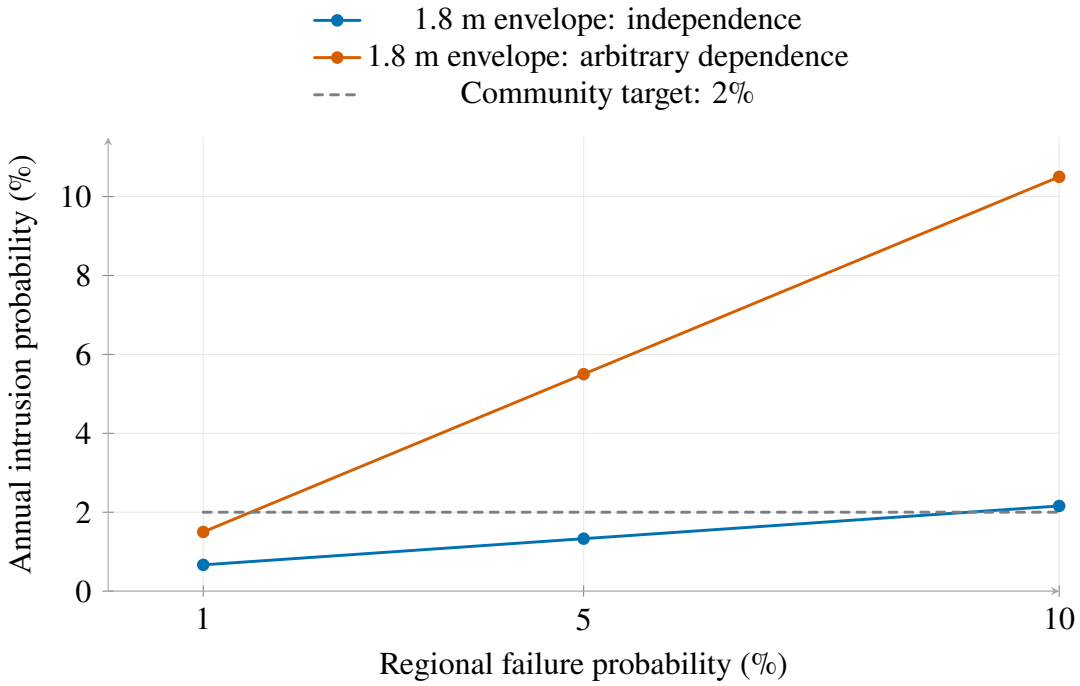


Figure 2. Failure dependence and preservation. The 0.30 m shift, local failure probability 0.5%, and 1.80 m envelope are fixed. The independence recommendation fails under attainable joint failure–weather distributions.

9 Insurance and a staged preservation decision

9.1 The restoration allowance is not the monument's value

We use a EUR 5 m restoration allowance B for an intrusion year, with an annual aggregate commitment of B. It is a funding scenario for a restoration campaign, not the Basilica's market value, an appraisal, or a fitted constant severity of actual claims. The insurer requires a claims-based schedule before offering an actual indemnity policy. Repeated events within a year share this allowance; costs beyond it remain the community's responsibility.

The single-landmark insurer has EUR 500,000 capital, EUR 20,000 administration and EUR 40,000 annual capital cost. Its reinsurance quote is $(1 - r)(1.25B\pi + 5000)$EUR. With a 20% premium expense share, its minimum premium is the maximum of the expected-cost and full-allowance solvency floors, just as in(7). We compare this with a EUR 200,000 annual coverage budget and inspect four retention shares.

The baseline failure scenario $f_L = 5\%$, $f_M = 1\%$ has worst annual intrusion probability 6% and a minimum EUR 540,000 premium. It cannot meet the declared budget. An audited local enhancement with $f_L = 0.5\%$ reduces the adverse probability to 1.5%, giving EUR 75,000 expected restoration funding and a EUR 195,469 premium at 10% retention. That pair leaves EUR 7,500 capital after the full EUR 5 m restoration commitment. The result links physical mitigation directly to insurance feasibility.

Table 7. Basilica preservation and insurance scenarios

Scenario	f_M	Worst risk	Premium (EUR)	Budget
Existing operations	1%	6.0%	540,000	Fails
Audited 1.4 m envelope	1%	1.5%	195,469	Passes
Audited 1.4 m envelope	5%	5.5%	501,719	Fails
Audited 1.8 m envelope	5%	5.5%	501,719	Fails

All rows use a 0.30 m shift and EUR 5 m annual restoration allowance. Failure rates and cost inputs are scenarios.

This feasible quote is fragile. Under the same price budget and 10% retention, the repair allowance cannot exceed EUR 5.095 m: the full-claim capital constraint binds before the expected-cost constraint. The annual intrusion probability cannot exceed 1.559% at $B = 5$m; hence with $f_L = 0.5\%$, the more restrictive insurance condition requires regional failure no greater than about 1.059%, rather than merely the 1.5% allowed by the preservation target. A EUR 10 m repair allowance reverses the decision even with the improved physical scenario. A different capital/reinsurance mix can change this boundary; our particular vector is not a universal requirement.

9.2 A renewal promise requires additional funding

The EUR 195,469 quote is single-year insurance. A full EUR 5 m claim leaves only EUR 7,500 capital, after which the same family cannot fit next year's quote to EUR 200,000. Under the 3-year capital agreement of Section 4.3, the local-upgrade scenario instead requires EUR 492,500 loss-absorbing capital each year and EUR 1,477,500 prefunded outside capital for all three annual restoration commitments. Its annual 2% fee is EUR 9,850, raising the quote to EUR 207,781.25. This fails the original EUR 200,000 annual budget. Even before changing other terms, fitting that original budget requires a commitment fee no greater than approximately 0.736%; no such offer has been observed.

An explicit expanded EUR 208,000 budget is a conditional alternative, leaving only EUR 218.75 annual headroom at the stated risk and cost inputs. Three consecutive full claims leave EUR 7,500 before each injection, and restore EUR 500,000 afterwards. The outside provider bears EUR 1,477,500 called capital, receives EUR 29,550 fees, and has EUR 1,447,950 net cash outflow on that path. It bears actual losses rather than receiving a fictional risk-free fee. The model does not establish its willingness or ability to supply that funding. This role could require a separately approved philanthropic or public reserve, or genuinely quoted loss-absorbing private equity; its funds are additional to project works and the insurer's existing capital. A repayable credit line must instead include debt service and may not preserve solvency.

All eight three-year zero/full-claim paths pass the revised solvency and commitment checks at the expanded budget. Smaller annual claims are covered by the monotone capital bound. No probability is assigned to those paths, no specific organisation is assumed committed, and no three-year agreement supports coverage beyond its expiry. If the original budget is fixed, maintain the one-year interpretation and prepare an explicitly funded restoration reserve; do not present EUR 195,469 as guaranteed continuing protection.

9.3 Recommended extent, costs, and decision gates

The immediate priority is an audit and operations package rather than a higher intrusive enclosure. We propose a EUR 250,000 initiation ceiling: surveying all relevant elevations and water paths, testing closure and backup power, documenting critical historic materials, and establishing an emergency conservation protocol. The community should seek procurement quotes for these work packages; the number is a proposed allocation, not a claimed tender price.

Reserve EUR 1.2 m for reversible local gaps revealed by the audit. The physical target is an effective 1.40 m envelope across entrances, rear paths and drainage, with monitored local annual failure no greater than 0.5%. Existing components meeting this criterion should be maintained, not replaced to spend the allocation. The operations ceiling is EUR 80,000/year. If survey or conservation approval rejects the candidate, the reserved construction money stays uncommitted and the action set is reopened.

At 30 years and 3%, the annual annuity factor is 19.600. Including the initial EUR 250,000, the programme's incremental present cost against the EUR 100,000/year baseline operations scenario is EUR 1.058 m. With adverse annual intrusion probability reduced by 4.5 percentage points, equation(14) gives a break-even total loss allowance of EUR 1.200 m per intrusion year. The proposed EUR 5 m repair scenario exceeds this threshold, so the preservation measure is economically justified within the declared scenario even before extra cultural loss is added. This physical-preservation calculation does not certify insurance renewal. The insurance decision must separately include outside funding and fees, and its annual budget must be expanded or its terms changed. Physical preservation remains possible even when continuing private coverage is unavailable.

We use four release gates. First, a conservation specialist approves the intervention's materials, reversibility and appearance. Second, surveyed hydraulic pathways demonstrate the effective threshold and MOSE's qualified success state. Third, operation and maintenance evidence supports the declared annual failure bounds; a few successful drills alone cannot establish a 0.5% annual failure probability. Fourth, a restoration cost assessment, signed reinsurance terms and an actually funded replenishment agreement fit the budget and capital inequalities for the promised coverage term. The evidence can come from engineering fault analysis and tested components as well as operating records, with uncertainty retained.

During the first three months, complete surveys, documentation and operating drills. Months four to six obtain conservation approval, detailed bids and a provisional flood/restoration insurance offer. Months seven to eighteen implement only approved gaps, commission independent leakage/power tests, and train conservators. Thereafter publish annual closure, leakage and maintenance records, review coverage each year, and revisit the design after any performance failure or material change in hazard. The timetable is a proposed implementation

plan, not evidence that those actions have happened.

If failures cannot be bounded near the insurance threshold, a public or philanthropic restoration reserve is preferable to pretending that the desired private premium is viable. The community may preserve an uninsurable landmark through transparent pooled funding, emergency collections protection, temporary closure, and a revised engineering plan. Insurance refusal is information about a financial promise, not a verdict that the Basilica has no value.

10 What is verified, and what remains to be measured

Thirty-eight independent checks accompany the current calculation. The verifier imports no model functions. It reconstructs yearly peak losses from the original NOAA series and primitive Venetian bins using scalar loops, then enumerates every enrollment/retention quote with 36-digit decimal cash flows. It independently recomputes the annual access stress and 30 discounted development payments. An eight-state linear program verifies the attainable dependence bounds and the minimum/maximum intrusion probabilities in 27 protection scenarios, propagating height-censoring intervals. Forty three-year stress paths test adaptive renewals, external capital calls and provider cash losses. Direct capital and demand checks verify the selected feasible witnesses. These checks establish computation within the specified models, not actual building performance.

Several broader claims are deliberately left unsupported. Eighteen years do not certify a 99% future return level. An empirical historical loss fraction is not a forecast of future claims. A margin challenge is not a calibrated confidence bound. Scenario damage slopes are not measured retrofit effects. A scenario project allocation is not a tender, and a scenario failure rate is not an achieved operational record. The strongest demonstrated results are the exact financial feasibility test over the stated finite domain, the attainable dependence counterexample, the three-year renewal failure and conditional funded alternative, and the linked development and preservation thresholds.

The most valuable next evidence is therefore specific. For a residential quote, obtain a surveyed floor and site transfer map, loss/repair records by depth and duration, an actual demand distribution, and reinsurance terms. For new development, add reachable service edges and allocation capacities. For the Basilica, measure all water paths and assess an authentic-material restoration campaign budget, then test a joint fault model under severe weather. These measurements target parameters that actually change the recommendations.

The framework answers the original dilemma with a disciplined rule: underwrite for a specified term when a jointly affordable, profitable and solvent contract exists with funded renewal; reward verified mitigation that expands that region; build only when the associated site and services pass; and preserve significance through explicit integrity and community-value conditions. The numerical demonstrations supply conditional decisions while retaining the missing evidence needed for real implementation.

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Letter to the Venice community

13 October 2026
To the residents of Venice, the Procuratoria di San Marco, and community representatives,

Dear neighbours and custodians,

Preserve St Mark's as a living place of worship and irreplaceable historic fabric. Insurance funds repair, not authenticity. Strengthen existing protection and close verified gaps before considering a larger enclosure.

Please authorise up to EUR 250,000 for the first three months: survey elevations and all water paths, test leakage and backup power, document historic materials, and prepare an emergency response. Existing glass needs maintenance and a rear/drainage exposure audit.

Reserve EUR 1.2 m for reversible gaps, released only after independent engineering and conservation approval. Our scenario requires an effective 1.40 m ZMPS envelope and local annual failure no greater than 0.5%; neither is established by current evidence. Obtain bids, approvals and insurance terms in months four to six; complete approved works and commissioning in months seven to eighteen. Allow up to EUR 80,000 annually for operations and conservation monitoring.

Under a stated 1% regional failure scenario, the worst annual intrusion opportunity falls from 6% to 1.5%. The EUR 5 m restoration allowance then has a EUR 195,469 single-year premium. That price fits the original EUR 200,000 budget, but a full claim leaves only EUR 7,500 insurer capital and cannot support the same renewal family next year.

Continuing protection therefore needs a separately funded capital agreement. At a hypothetical 2% annual commitment fee, EUR 492,500 must be available for loss-absorbing injections each year: EUR 1,477,500 prefunded over three years, additional to the works and the insurer's existing EUR 500,000. The provider bears these called losses without repayment; this is not an assumed free loan. Its EUR 9,850 annual fee raises the premium to EUR 207,781, so the original budget fails. Consider an explicit EUR 208,000 coverage budget only if a real provider, restoration assessment and signed reinsurance terms support it. The model does not establish that such terms are available, and three years of funds do not guarantee coverage after expiry.

Do not approve a higher wall by assuming failures and extreme tides are independent. Our separate calculation shows that this can reverse the decision. Publish annual failure records and reassess after an intrusion or material risk change.

The proposed EUR 1.45 m works budget excludes annual costs and outside capital and requires actual bids. If performance, conservation or financing cannot be verified, keep construction funds uncommitted, protect movable collections and continuity of worship, and reopen the options. Preserving St Mark's may require transparent public or philanthropic restoration funding even where continuing private insurance is unavailable.

Respectfully,
Insurance of Catastrophes modelling team

Report on Use of AI

This report was prepared with OpenAI Codex, a GPT-6-based autonomous assistant, on 13 October 2026. The exact served model identifier and complete host/API message log were not available. The assistant selected the route and objects, retrieved data, authored code and text, and performed author-side checks. An external coordinating assistant supplied task and workflow boundaries, then two directed feedback rounds: renewal closure and height censoring, followed by a local correction to the unchanged-risk explanation. The author independently implemented and checked these revisions. AI checks are not human validation.

The scope includes model selection, financial and preservation derivation, data ingestion, independent numerical implementation, LaTeX generation and language editing. All source facts used for parameters were checked against the cited first-party materials. Financial costs, vulnerability, demand, failure and project allocation parameters remain explicitly labelled scenarios. Code execution used Python 3.12, NumPy 2.5.3 and SciPy 1.18.1 under the fixed Praxis 0.7.1 source; xlrd 2.0.2 was isolated to this case for source workbook ingestion. The first complete 19-page and initial-final 20-page drafts remain preserved. Revisions add censored-height intervals and the cost and expiry of renewal capital.

The same-problem student solutions, award papers, code and commentary were not consulted. Pretraining exposure is unknown, and this is not claimed to be a technically isolated blind test.

Task input retained in the available record. The coordinator requested an independent complete 2024 ICM-E paper covering underwriting, two continents, development, preservation, a landmark other than Cape Hatteras, and a one-page letter, with a fixed workflow/budget and preserved runs. Same-problem solutions and other cases were excluded. The four-page original was read completely. This is a task summary, not a verbatim export.

Output and verification record. The full manuscript and source code are the actual AI-authored outputs. The current computation is run20261012T232719-50dd3573, with 38 named independent implementation checks. All six run attempts are retained; five passed and one failed. Runs 1–4 established the original calculations and challenges; their complete records remain preserved. Attempt 5 model execution completed but its validator failed on an unhandled empty renewal quote; the source, output and error are retained. Attempt 6 repaired that validator branch and verified the genuine rejected cases. The first directed science-feedback round added three-year renewal/funding checks, fee-inclusive NPV and censored-height probability intervals. The second round corrected only that explanation; code, data, run 6 and numerical results stayed unchanged, with no new model run. Neither checks nor compilation establish calibration or award level.

Unavailable detail. The complete raw conversation, exact model metering, and complete intermediate AI tool outputs were not exposed for export. No reconstructed conversation is presented as the full record requested by COMAP's 2024 examples. A formal contestant submission would have to retain and provide that original record as required. The document's internal control identifier 2024022 is not an official team allocation. These disclosure limits are recorded rather than concealed.